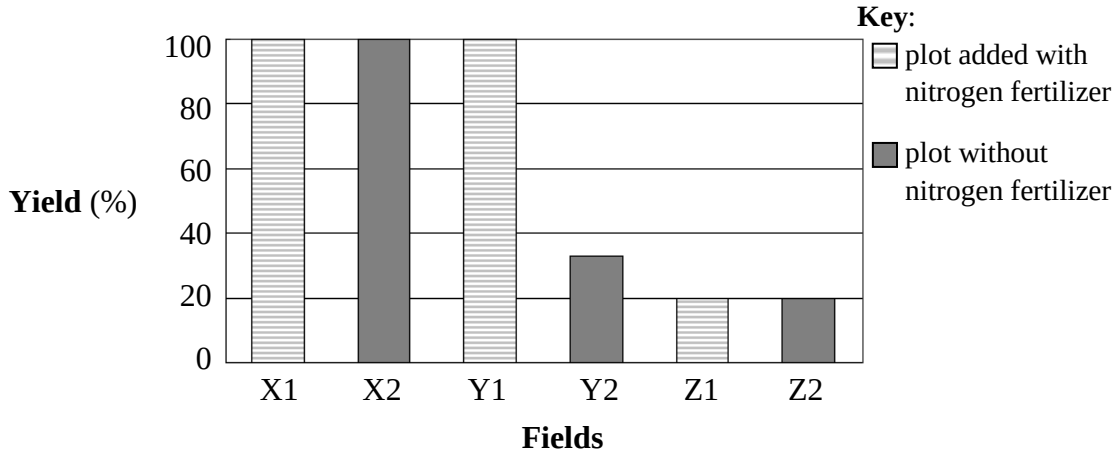
Structured questions

|!|BCELA0100000001|!| ★★

The same type of wheat was grown in three fields (X, Y and Z). The soil conditions in the three fields were the same, except for nitrogen composition. An experiment was carried out to investigate the growth of wheat with or without nitrogen fertilizers. Each field was divided into two plots. One plot was added with sufficient nitrogen fertilizer, while the other plot was without nitrogen fertilizer. After 20 weeks of growth, wheat in the three fields was harvested. Their yields are recorded below.



(a) Describe and explain ***one*** symptom of nitrogen deficiency in plants. (2 marks)

(b) Which field(s) required the addition of nitrogen fertilizer to achieve a maximum yield? Explain your answer. (3 marks)

(c) Describe and explain the results of the other field(s). (4 marks)

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##

(a) Plants deficient in nitrogen develop yellow leaves (1)
because nitrogen is required for the synthesis of chlorophyll. (1)
OR
Plants deficient in nitrogen display reduced growth (1)
because the plants do not have enough nitrogen to synthesize amino acids and nuclei acids which are essential for making new cells. (1)

(b) Field Y1. (1)
The addition of nitrogen fertilizer led to an increase in yield to reach the maximum level compared with field Y2 (the control). (1)
This increase is not observed in the other two fields. (1)

(c) In field X, nitrogen is not deficient. (1)
The plot without the addition of nitrogen had reached maximum yield. / The addition of nitrogen did not further increase the yield compared to the control. (1)
In field Z, nitrogen is not deficient. (1)
The addition of nitrogen did not increase yield of plot Z1 compared to Z2 (the control). (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Analysing data	3
(c)	Analysing data	4

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|!|BCELB0100000002|!| ★ **SI** **Cross topic**

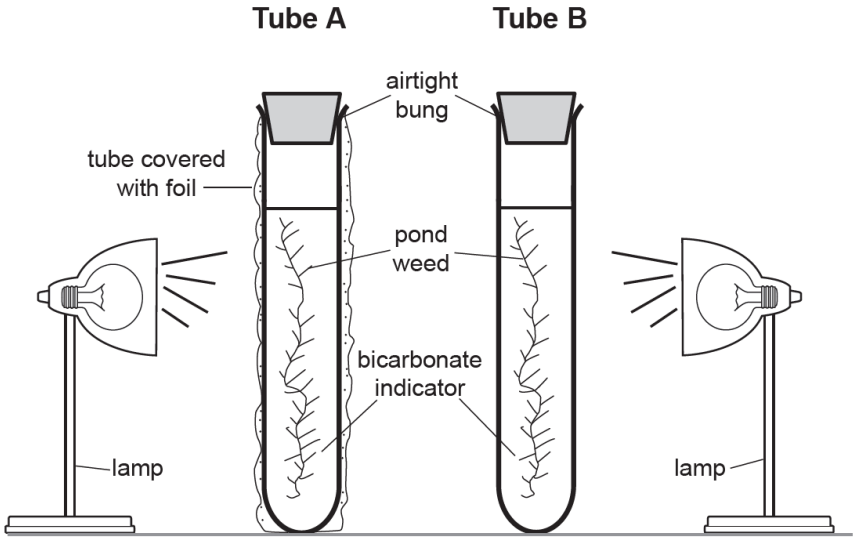
CCEA GCE AS Biology Assessment Unit AS 2 Jun 2016 Q2

(a) In mammals, the gas exchange surface is provided by the lung alveoli. In plants, the spongy mesophyll of the leaves performs this role.

(i) Identify ***one*** other difference between gas exchange in mammals and plants. (1 mark)

(ii) Describe ***one*** way in which a diffusion gradient is maintained in the mammalian lungs. (1 mark)

(b) An investigation of gas exchange in plants was carried out, using bicarbonate indicator. The apparatus used is shown in the diagram below.



(i) Complete the table below to show the expected colour of the bicarbonate indicator in each tube after three hours in bright light. (1 mark)

Tube	Colour of bicarbonate indicator	
	At start	After three hours
A	red	
B	red	

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(ii) Explain fully the result for Tube B. (2 marks)

(c) Describe how you would extend the investigation to demonstrate the compensation point of the pondweed. (3 marks)

##

(a) (i) In mammals, gas exchange involves an active ventilation process; In plants, there is no active ventilation process.
OR
In mammals, gas exchange is only linked to respiration; In plants, both respiration and photosynthesis are involved. (1)

(ii) Ventilation bringing in air rich in oxygen / removing air with higher levels of carbon dioxide / blood flow ensures concentration gradient maintained by removing oxygen / returning carbon dioxide to the lungs (1)

(b) (i) A – yellow; B – purple (1)

(ii) The rate of photosynthesis exceeds the rate of respiration, (1)
leading to a reduction of carbon dioxide in the water. (1)
This causes the indicator to turn purple.

(c) Move the lamp / use a lamp with a dimmer switch to adjust light intensities / use different layers of muslin. (1)
Increase (or decrease) light intensities until bicarbonate indicator remains red / shows no change after a period of time. (1)
Control other variables, e.g. room temperature, use the same pondweed each time (1)

(8 marks)

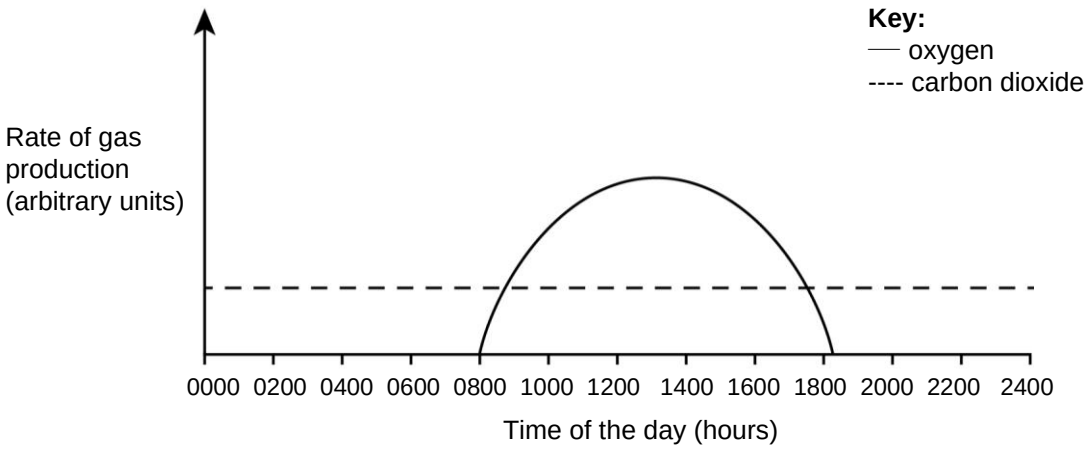
##

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	Skills assessed	Marks
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	1
(b)(i)	Applying concepts	1
(b)(ii)	Applying concepts	2
(c)	Practical skills	3

|!|BCELB0100000003|!| ★

The graph below shows the rates of oxygen production and carbon dioxide production of a plant on a winter day.



(a) State the times at which there is no net exchange of gases into or out of the plant. (2 marks)

(b) Sketch a line in the graph above to show the oxygen production rate of the plant on a summer day. (2 marks)

(c) The area below the solid line (line showing the oxygen production rate) is smaller than the area below the dotted line (line showing the carbon dioxide production rate). Describe the condition of the plant and suggest how the plant can survive the winter. (4 marks)

##

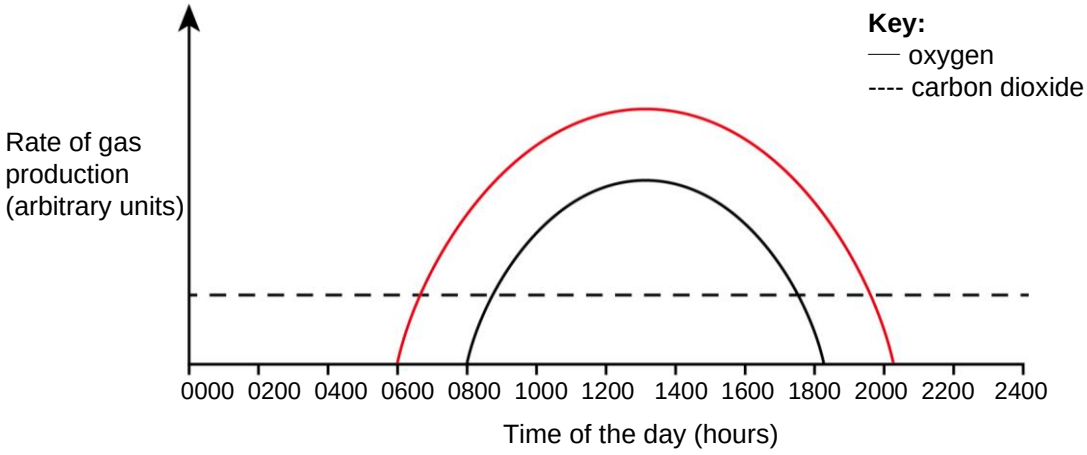
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(a) 0900 (1)

1700 / 1800 (1)

(b) Correct drawing:

- Longer light period (1)
- Higher overall rate (1)



(c) The area below the solid line represents the food produced in 24 hours. (1)

The area below the dotted line represents the food consumed in 24 hours. (1)

Food production does not meet the demand for food. (1)

The plant may enter dormancy / utilize some of the stored food / shed some leaves to lower energy expenditure. (1)

(8 marks)

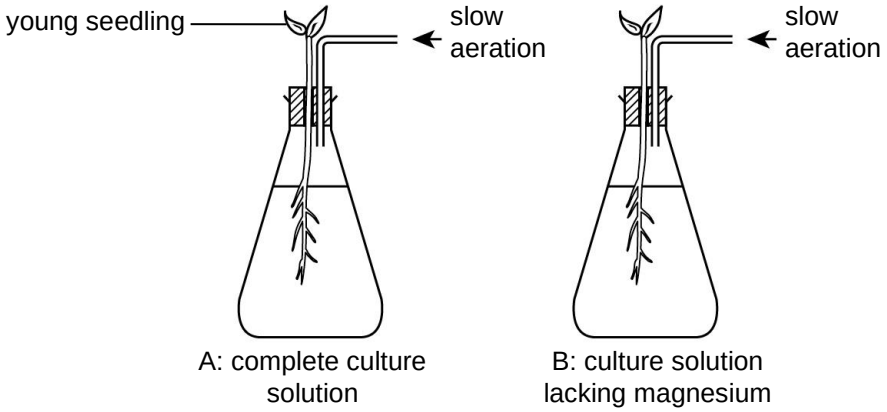
##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Presenting information	2
(c)	Analysing data	3
	Applying concepts	1

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|!|BCELB0100000004|!| ★ **SI**

To investigate the effects of magnesium on plant growth, a student set up a water culture experiment, as shown in the diagram below.



(a) (i) The apparatus was not set up properly. Suggest, with reasons, ***two*** corrections that you would make to ensure the experiment can be carried out properly. (4 marks)

(ii) After your corrections, what would be the appearance of the seedling in flask B at the end of the experiment? (1 mark)

(b) Besides magnesium ions, suggest ***one*** mineral ion that may be present in the complete culture solution. State ***one*** importance of this mineral to plant growth. (2 marks)

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(c) Suggest ***two*** control variables of the experiment. (2 marks)

_____

_____

_____

##

(a) (i) The flasks should be wrapped with aluminium foil. (1)
This can prevent the growth of algae otherwise the absorption of minerals will be affected. (1)
The glass tubes should dip into the solution (1)
so that oxygen / air can be supplied to root cells for respiration. (1)
(ii) The leaves would turn yellow. (1)

(b) Nitrates (1)
Nitrogen is essential to the synthesis of amino acids / proteins. (1)
(Or other acceptable answers)

(c) The two young seedlings should be of the same species / of the same growing stage / of a similar length. (1)
The two set-ups should be placed under the same light intensity. (1)
(Or other acceptable answers)

(9 marks)

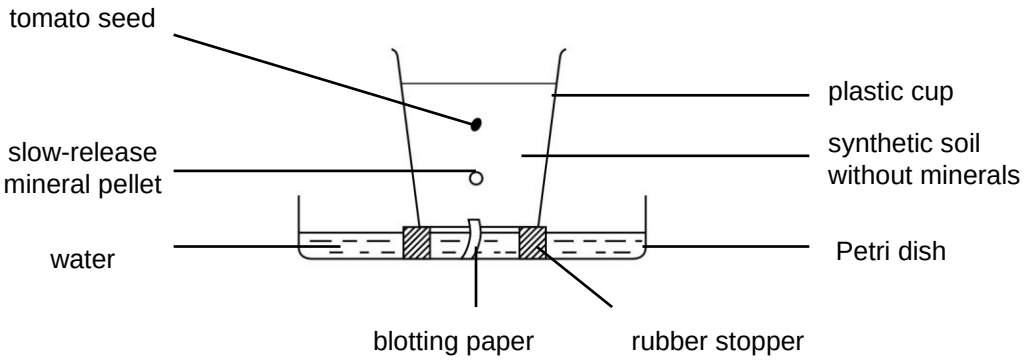
##

	Skills assessed	Marks
(a)(i)	Practical skills	4
(a)(ii)	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)	Practical skills	2

|!|BCELB0100000005|!| ★ **SI**

A student carried out an experiment to investigate the effects of different levels of minerals on the growth of the roots of tomato plants. She set up eight apparatuses (set-ups 1–8) and one of these apparatuses is shown below. She varied the number of slow-release mineral pellets used per plastic cup. After three weeks, she measured the fresh mass of each tomato root. She then obtained results from other classmates who had carried out the same investigation to obtain averages. The results are shown in the table below.

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Set-up	Number of slow-release mineral pellets used	Average fresh mass of tomato root (g)
1	2	5.0
2	4	11.1
3	6	16.9
4	8	21.1
5	10	24.1
6	12	24.5
7	14	24.2
8	16	24.0

(a) Plot a graph to show the results of the experiment. (4 marks)

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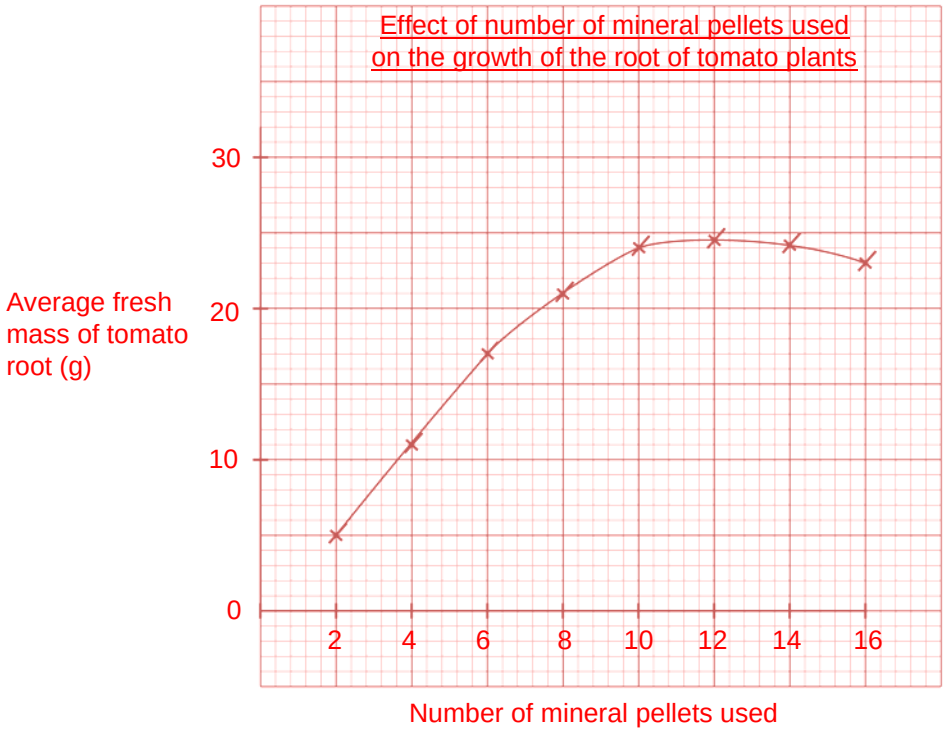
(b) With reference to the graph, describe the results of experiment. (2 marks)

(c) What is the importance of using mineral-free synthetic soil in this experiment? (1 mark)

(d) What procedure was carried out by the student to increase the reliability of the results? (1 mark)

##

(a) Correct title (1)
Correct choice of axes (1)
Axes with labels and units (1)
Correct plotting and joining of lines (1)



(b) The average fresh mass of tomato root significantly increases with the increase in the number of mineral pellets used up to 10. (1)
Further increase in the number of mineral pellets does not increase the growth of tomato root. (1)

(c) To ensure that only minerals available to the plant are those supplied by the pellets. / If normal soil is used, the minerals present in the soil will affect the results. (1)

(d) The student collected results from other classmates and calculated average. (1)

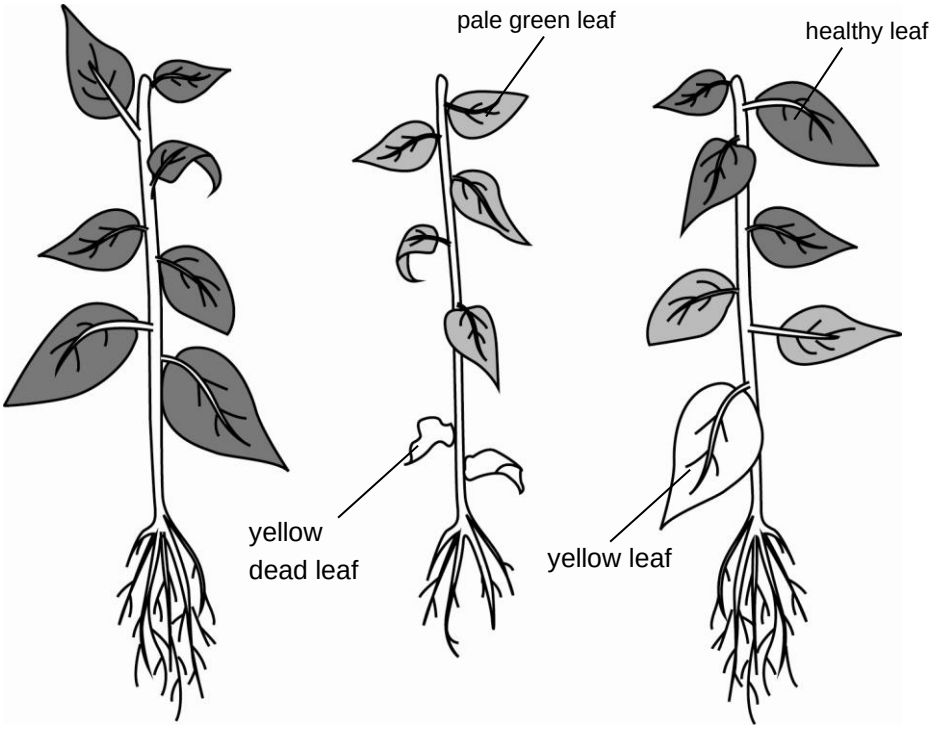
(8 marks)

##

	Skills assessed	Marks
(a)	Presenting information	4
(b)	Analysing data	2
(c)	Practical skills	1
(d)	Practical skills	1

|!|BCELB0100000006|!| ★ **SI**

Katie performed an experiment with three healthy plants to study the effects of nitrate and magnesium deficiencies on the plants' growth. She kept the plants under the same conditions, except for the types of minerals supplied in the soil. The following diagrams show the results after four weeks.



(a) State ***two*** variables, other than the concentration of mineral ions, which should be kept constant for all the plants. (2 marks)

(b) Compare plants X and Y.

(i) State ***two*** differences in their appearances. (2 marks)

(ii) Explain these differences. (2 marks)

(c) Compare plants X and Z.

(i) State ***one*** difference in their appearances. (1 mark)

(ii) Explain this difference. (1 mark)

(d) If the soil is deficient of nitrate and magnesium, crops will not grow well. State ***one*** method that a farmer can use to improve the mineral content of the soil. (1 mark)

##

(a) Any ***two*** of the following: (2)

- Light intensity (1)
- Temperature (1)
- Humidity (1)
- Carbon dioxide level (1)
- Soil type (1)

(Or other reasonable answers)

(b) (i) Any ***two*** of the following: (2)

- The leaves of plant Y are smaller and paler than that of plant X. / The lower leaves in plant Y is yellow or dead while the leaves of plant X is healthy. (1)
- The stem of plant Y is thinner than that of plant X. (1)
- The root development of plant Y is poorer than that of plant X. (1)

(ii) There is not enough nitrate ions for synthesizing proteins, (1)
which are needed to make new cells. (1)

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(c) (i) Plant Z has paler leaves. / Yellow leaves are formed in plant Z while there are no yellow leaves in plant X. (1)

(ii) Magnesium is necessary for making chlorophyll molecules. (1)

(d) Any ***one*** of the following: (1)

- Addition of fertilizer (1)
- Planting of leguminous plants (1)

(Or other reasonable answer)

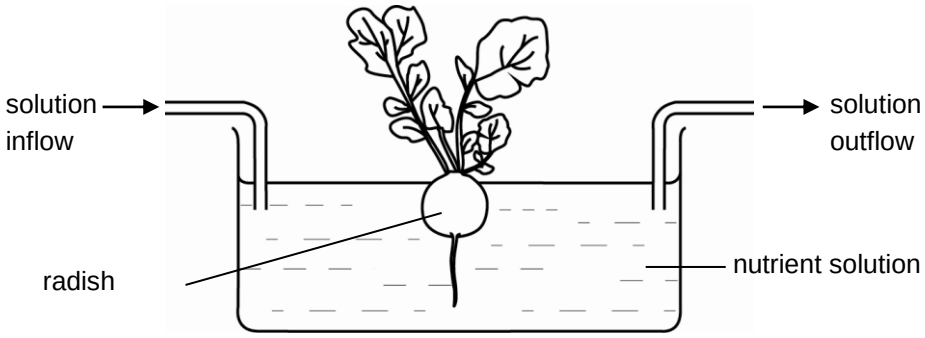
(9 marks)

##

	Skills assessed	Marks
(a)	Practical skills	2
(b)(i)	Analysing data	2
(b)(ii)	Applying concepts	2
(c)(i)	Analysing data	1
(c)(ii)	Applying concepts	1
(d)	Understanding basic concepts	1

|!|BCELB0100000007|!| ★ **SI**

Betty carried out an investigation to study the growth rate of radish plants by a method called hydroponics. The following shows the set-up.



(a) State ***two*** factors that Betty should keep constant during the investigation.

(2 marks)

(b) Explain why the solution flowing into the nutrient solution should be aerated.

(3 marks)

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(c) Betty measured the total dry mass of five radish plants at one-week interval over six weeks. The following table shows the results.

Time (week)	Total dry mass of five radish plants (g)
0 (start)	0.25
1	0.65
2	3.10
3	8.75
4	10.20
5	13.35
6	14.00

(i) Describe the changes in the dry mass of the radish plants during the six weeks.
(3 marks)

(ii) Based on the relationship between photosynthesis and respiration, state the cause for the change in the dry mass of the plants during the six weeks.
(1 mark)

##

(a) Any ***two*** of the following: (2)
- light intensity (1)
- temperature (1)
- humidity (1)
- carbon dioxide level (1)
- wind speed (1)
- volume of nutrient solution (1)
- plant species (1)

(Or other reasonable answers)

(b) Oxygen can support respiration (1)
that releases energy for the uptake of mineral ions by active transport. (1)
The mineral ions are essential for building body tissues / regulating metabolism in plants. (1)

(c) (i) The dry mass of radish plants increased steadily during week 1 and week 2 (1)
The dry mass increased quickly during week 3 (1)
and then increased steadily after week 3. (1)

(ii) When the rate of photosynthesis is higher than the rate of respiration, the dry mass increases. (1)

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(9 marks)

##

	Skills assessed	Marks
(a)	Practical skills	2
(b)	Practical skills	3
(c)(i)	Analysing data	3
(c)(ii)	Applying concepts	1

|!|BCELA0100000008|!| ★★

Hoagland was a plant scientist who carried out an investigation to measure the concentrations of mineral ions in the cell sap of a freshwater plant and the surrounding pond water. His results are shown in the following table.

Site of measurement	Concentrations of ions (mg / L)			
	Sodium	Magnesium	Potassium	Chloride
Cell sap	1980	260	2400	3750
Pond water	28	360	2	35

(a) With reference to the table, state ***one*** ion that could have entered the plant mainly by diffusion. Explain your answer. (3 marks)

(b) Hoagland found that some ions were taken up by the plant only if oxygen was bubbled through the pond water. Explain his observation. (3 marks)

(c) Besides mineral ions, name ***two*** other environmental factors the freshwater plant needs for growth and explain why these factors are necessary. (4 marks)

##

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(a) Magnesium ion (1)

This is because the concentration of magnesium ion is higher in pond water than that inside the plant cells. (1)

So, magnesium ions can diffuse along the concentration gradient into the cells. (1)

(b) Since some ions are taken up into plants by active transport (1)

and active transport requires energy released by respiration. (1)

More energy can be released from cells through respiration when oxygen is sufficient. (1)

(c) Any ***two*** sets of the following: (4)

1. Light intensity (1)

Light supplies energy for the plant to carry out photosynthesis to produce food. (1)

2. Carbon dioxide level (1)

Carbon dioxide is required by the plant as the raw material in photosynthesis. (1)

3. Temperature (1)

An optimum temperature can facilitate the actions of enzymes that involve in photosynthesis. (1)

(10 marks)

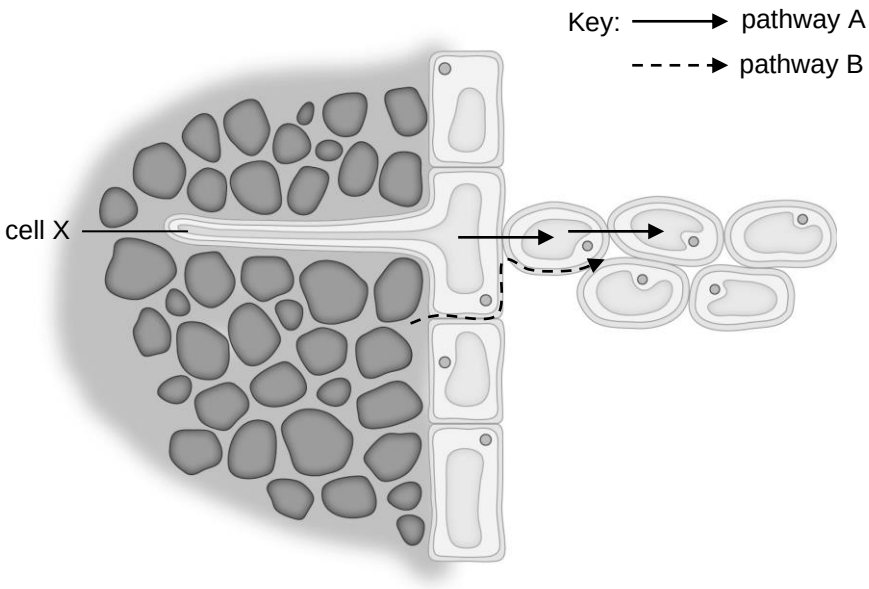
##

	Skills assessed	Marks
(a)	Analysing data	1
	Applying concepts	2
(b)	Applying concepts	3
(c)	Understanding basic concepts	4

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|!|BCELB0100000009|!| ★

The following diagram shows the root cells of a plant and the surrounding soil particles.



(a) State ***two*** structural differences between cell X and a leaf mesophyll cell. (2 marks)

(b) Name the mineral ions absorbed by the roots that are essential components of

(i) nucleic acids, and (1 mark)

(ii) chlorophyll. (1 mark)

(c) Pathways A and B are two main transport pathways of mineral ions in the root.

(i) Describe and explain how waterlogged soil affects pathway A. (3 marks)

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(ii) If the mineral concentration in the soil is high, explain why pathway B is more suited to the uptake of minerals. (3 marks)

##

(a) Any ***two*** sets of the following comparisons: (2)

Cell X	Leaf mesophyll cell
- develops long and fine root hairs	- has no projections on the surface
- contains no chloroplasts	- contains many chloroplasts
- not cylindrical in shape	- cylindrical in shape

(Or other acceptable answers)

(b) (i) Phosphate / nitrate (1)

(ii) Magnesium / nitrate (1)

(c) (i) Waterlogged soil is poorly aerated; the plant's root cells cannot obtain enough oxygen for respiration. (1)

Thus less mineral ions can be absorbed from the soil by active transport (1)

that requires energy from respiration.. (1)

So, less mineral ions are absorbed through pathway A.

(ii) The high concentration of mineral ions in soil forms a concentration gradient, (1)

which leads to the diffusion of mineral ions (1)

across the fully permeable cell wall through pathway B. (1)

(10 marks)

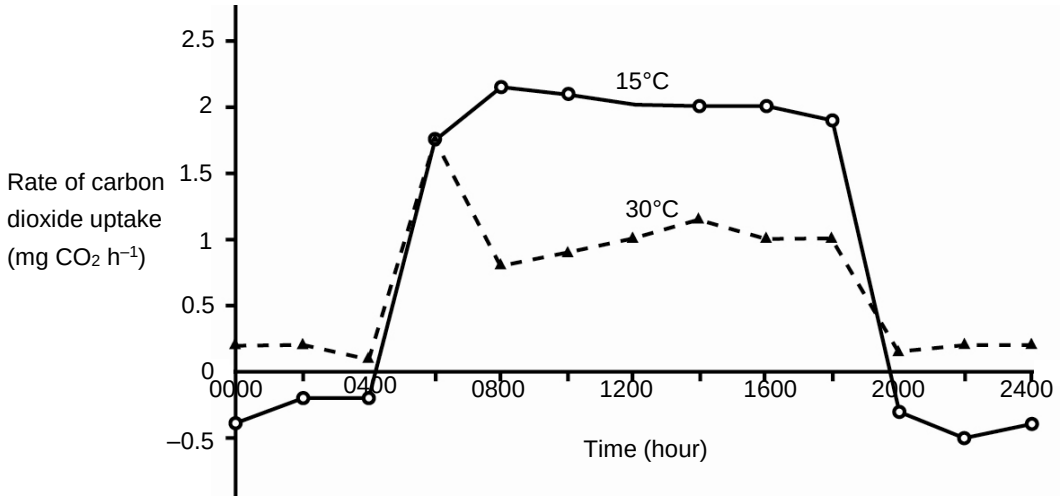
##

	Skills assessed	Marks
(a)	Analysing data	2
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	1
(c)(i)	Applying concepts	3
(c)(ii)	Applying concepts	3

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|!|BCELA0100000010|!| ★★

The following graph shows the rate of carbon dioxide uptake by a plant at 15 °C and 30 °C during a 24-hour period.



(a) State the times at which the carbon dioxide uptake is the same at both temperatures. (1 mark)

_____

(b) Compare the rate of carbon dioxide uptake at 15 °C and 30 °C in daylight and darkness. (3 marks)

_____

_____

_____

_____

(c) State ***one*** possible reason to explain the difference in the carbon dioxide uptake at both temperatures during the daytime. (1 mark)

_____

_____

(d) (i) Given that Area A is the area below the curve and above the x-axis; and Area A can represent the amount of food production through photosynthesis. What can Area B, the area above the curve and below the x-axis represent? (1 mark)

_____

_____

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(ii) Based on the areas mentioned in (i), how can we determine the net food production by the plant? (1 mark)

(iii) In terms of the net food production by the plant, which temperature is the most suitable for the growth of the plant? Explain why. (3 marks)

##

(a) 06:00, 19:30 (1)

(b) There is a greater uptake of carbon dioxide at 15 °C than 30 °C during the hours of daytime. (1)

There is a greater uptake of carbon dioxide at 30 °C than 15 °C during the hours of darkness. (1)

The uptake of carbon dioxide at 30 °C is always positive while that at 15 °C becomes negative at night time. (1)

(c) Enzymes may be denatured at high temperature and this inhibits the rate of photosynthesis. (1)

(d) (i) It can represent the amount of food consumed by respiration. (1)

(ii) Net food production by the plant = Area A – Area B (1)

(iii) 15 °C (1)

Since the net food production by the plant at 15 °C is greater than that at 30 °C (1)

the growth rate by the plant is greater at 15 °C than that at 30 °C. (1)

(10 marks)

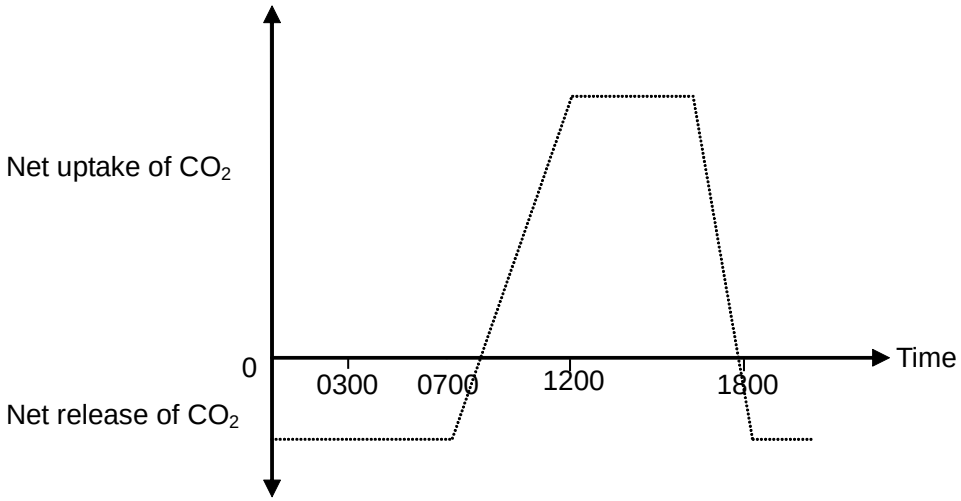
##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	3
(c)	Applying concepts	1
(d)(i)	Applying concepts	1
(d)(ii)	Applying concepts	1
(d)(iii)	Analysing data	1
	Applying concepts	2

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|!|BCELC0100000011|!|

The graph below shows the net uptake and release of carbon dioxide of a local plant during a day.



(a) Explain for the shape of the curve during the time from 03:00 to 07:00. (3 marks)

(b) (i) State the time period during which the curve levels off. (1 mark)

(ii) Why does the graph level off? (1 mark)

(c) (i) Suggest ***one*** environmental change that can lower the height of the curve's peak. (1 mark)

(ii) Describe how the condition you answered in (c)(i) causes the change. (2 marks)

##

(a) During darkness, no photosynthesis occurs (1)
and only respiration takes place, (1)

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so there is a net release of carbon dioxide by respiration. (1)

(b) (i) 12:00 to 17:00 (1)

(ii) This is because other factors (e.g. carbon dioxide concentration) limit the rate of photosynthesis. (1)

(c) (i) decrease in temperature / decrease in light intensity / other reasonable answer(1)

(ii) Decrease in temperature can decrease the activity of enzymes involved in the reactions of photosynthesis.

OR

Decrease in light intensity can decrease the amount of light energy used to start the photosynthetic process. (1)

Therefore, the photosynthetic rate decreases and this decreases the uptake of carbon dioxide. (1)

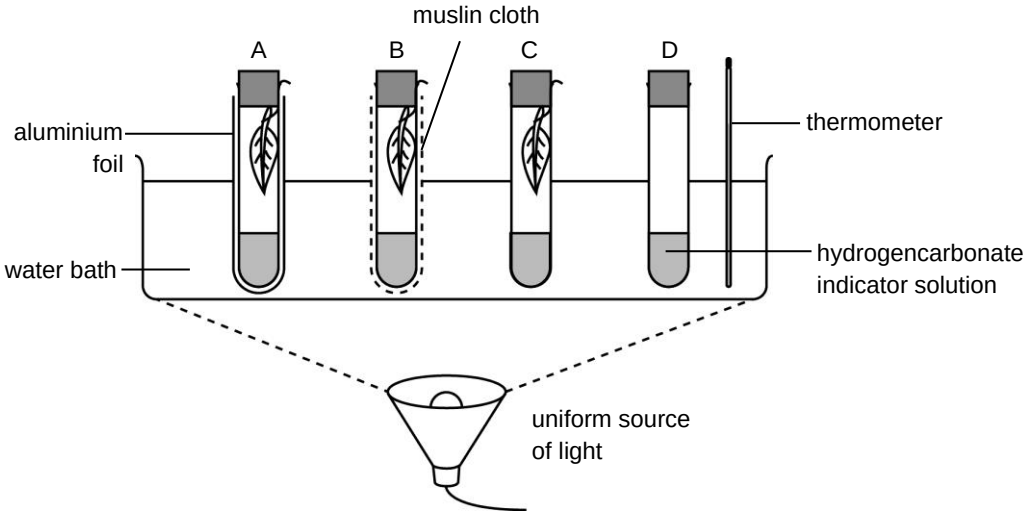
(8 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
	Applying concepts	2
(b)(i)	Analysing data	1
(b)(ii)	Applying concepts	1
(c)(i)	Applying concepts	1
(c)(ii)	Applying concepts	2

|!|BCELB0100000012|!| ★ **SI**

The following diagram shows an experiment used to investigate the gas exchange in green plants. The hydrogencarbonate indicator solutions in test tubes A to D were red at the beginning of the experiment.



(a) What is the purpose of setting up test tube D in this experiment? (2 marks)

(b) What is the purpose of immersing the test tubes in a water bath? (1 mark)

(c) Suggest ***one*** variable that should be kept constant during the investigation. (1 mark)

(d) The following table shows the results of the investigation after three hours.

Test tube	Final colour of the indicator solution
A	yellow
B	red
C	purple
D	red

(i) Explain the change in the indicator solution in tube A. (2 marks)

(ii) Explain the change in the indicator solution in tube C. (2 marks)

(iii) Explain why there is no change in the indicator solution in tube B. (2 marks)

##

(a) Test tube D serves as a control (1)
It is used to see if colour change of indicator solution is due to the presence of leaf. (1)

(b) To provide a constant temperature to all test tubes (1)

(c) Carbon dioxide level / size of the leaf / concentration of hydrogencarbonate indicator solution (1)
(Or other reasonable answers)

(d) (i) In darkness, respiration occurs and photosynthesis does not occur in the leaf.

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(1)

More carbon dioxide is thus released. (1)

(ii) Under bright light, the rate of photosynthesis is higher in the leaf. (1)

More carbon dioxide is absorbed by the leaf. (1)

(iii) The rate of respiration equals to the rate of photosynthesis. (1)

The uptake and release of carbon dioxide by the leaf is balanced. (1)

(10 marks)

##

	Skills assessed	Marks
(a)	Practical skills	2
(b)	Practical skills	1
(c)	Practical skills	1
(d)(i)	Applying concepts	2
(d)(ii)	Applying concepts	2
(d)(iii)	Applying concepts	2

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